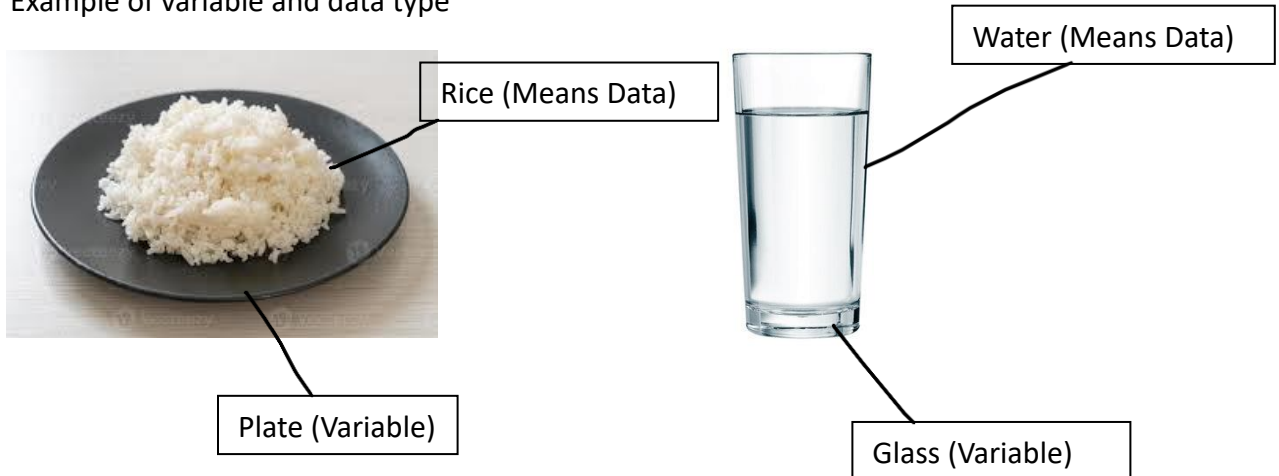


Variable (Let, Const)

1. Variable
2. Data Type (Number, String, Boolean, Null, Undefined)

Example of variable and data type



Variable is just like a container which store different type of data. Variable को types हुन्छ । त्यो हो Data Types ES6 Develop from 2014.

Data Types :

```
// String "Dipkiran" 'dipkiran' '1'
```

```
let name = "DipKiran"
```

```
// Number 1 2 3 4 5.1 6.7 8.0
```

```
//1 2 3 4 integer number
```

```
//2.3 4.5 float number
```

```
let a= 5; //integer number
```

```
let b = 1.5; // float number
```

```
// boolean true false
```

```
let flag = false;
```

```
//undefined
```

```
let country;
```

```
//null
```

```
let city = null;
```

❓ let = **keyword** (is a keyword used to declare a variable)

❓ name = **variable** (identifier / variable name)

❓ "dipkiran" = **value** (specifically, a string literal)

```
//object {}
```

```
let person {
```

```
  name : "dipkiran" //string
```

```
  address : "Dharan" // string
```

```
  number : 123333 // number
```

```
  age : 42 //number
```

```
}
```

Variables Let / Const /

Const Value cannot be changed

Arrey :- भनेको एउटा भोला हो भनेर सम्झनु । जहाँ एउटै किसिमको डाटाहरु कलेक्सन हुन्छ । जस्तै चिनीको व्याग भित्र चिनी मात्र हुन्छ, हरु मिसाएको हुदैन । **Collection of Same type of Data.** Array लाई [] यसरी लेखिन्छ ।

```
let jhola = [1,2,3,4,5];
```

Operators

Arithmetic Operator भित्र के के पछि भन्दा (+ - / * %) कुनै

पनि हिसाब किताब गर्नु परो भने Arithmetic Operator को प्रयोग गरिन्छ ।

```
let a = 5;
let b = 10;
let c = (a+b);
console.log (c);
```

Assignment Operator

+= -= /= *= %=

```
let a = 9;
a += 2;
console.log (a);
Result (11)
```

- a += 5; यसरी लेख्दा हामिले a को value 5 लेखेको हुनाले 9+5 जोडेर यसको Result 11 आउछ ।
//a = a + 5

Comparison (Relational Operator)

Used to compare values → result is true or false.

> < == >= <= != ===

(> Greater then) (< less then) (== equal to) (>= Greater or Equal) (<= Less or Equal) (!= Not Equal) (=== Aject Equal)

Relational Operator जहिले पनि हामि condition मा use गर्छौ । condition लाउदा खेरी जस्तै If, while, तिर Relational Operator लाई apply गर्छौ ।

Conditions

- if, else, else if

```
let a = 9;
if (a == 10){
  console.log ("10 is Greater then 9");
}
else{
  console.log ("Now Equal");
}
```

उदाहरण : यदि ९ र १० बराबर हो भने "Print Equal" यदि Value बराबर होइन भने "Print Not Equal".

Logical Operator

(&& and) (|| or) (! not)

Logical Operator चाहिए हमारे User Name or Password अथवा Form Validation गर्न को लागि प्रयोग गर्छौ ।

```
let username = "dipkiran1";
let password = "123";

if (username == "dipkiran" && password == "123"){
  console.log ("Login Sucessfull");
}

else{
  console.log("Login Fail");
}
```

Increment and Decrement

increment ++

decrement --

```
let a= 10;
console.log (++a);
console.log (a++);
```

If / Else

केही कुराहरुलाई compare गर्नु परो भने If अथवा Else को प्रयोग गरीन्छ ।

```
let keyTimiLaiBhokLago = "";
if(keyTimiLaiBhokLago == "yes"){
  console.log ("Babu timi khana khau");
}
else{
  console.log("xoina vhane nacha");
}
```

Switch

```
let day = 3;

switch(day){
  case 1:
    console.log("Sunday");
    break;

  case 2:
    console.log ("Monday");
    break;

  case 3:
    console.log("Tudsdays");
    break;

  default:
    console.log("there are 03 days in a week");
}
```

Loops

- for, while

03 Types of Loop

While Loop जब सम्म मेरो थिर्खा भेटिदैन तब सम्म पानि पिउछु ।

For Loop

Do While

एउटै कुरालाई Repet गरेर देखाउनु लाई Loop भनिन्छ । पहिले हामिले `hello world` लाई २ चोटी `print` गर्नु लाई २ चोटी `console.log("Hello World");` लेख्नु पर्छ्यो । यदि Hello World लाई १०० चोटी प्रिन्ट गर्ने पर्ने भयो भने १०० चोटी `console.log("Hello World");` लेख्नु पर्छ्यो । अब While Loop को प्रयोग गरेर Hello world लाई एउटै Code मा १०० चोटी देखाउन सकिन्छ ।

//while loop - jaba sama

```
let a = 1;
while(a <= 10){
  console.log(a);
  a++;
}
```

```
let count = 1;
while (count <= 20){
  console.log(count + "..Dipkiran");
  count++;
}
```

```
let a = 1;
while(a <= 10){
  if(a % 2 == 0){
    console.log(a);
  }
  a++;
}
```

//For loop

```
for (a=1; a <= 10; a++){
  console.log(a);
}
```

1. What JavaScript is & where it runs (browser)
2. Variables
 - `let`, `const`
3. Data types
 - Number, String, Boolean, Null, Undefined
4. Operators
 - `+` `-` `*` `/` `%`
5. Conditions
 - `if`, `else`, `else if`
6. Loops
 - `for`, `while`
7. Functions
 - normal functions & arrow functions

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Task.

- if the marks is above 60 print A Grade if the mark is above 80 print A+ if the mark is less then 60 print fail using relation operator.
- If 9 is greater 10 then print A is greater using relation operator.
- User name and password wrong using logical operator.

-

✅ Task 1 Solution

Give 20% discount if cartTotal $\geq$ 2000, otherwise 10% discount if member

```
js                                                                    Copy code

let cartTotal = 2500;
let isMember = true;
let discount = 0;

if (cartTotal >= 2000) {
  discount = 20;
} else if (isMember) {
  discount = 10;
}

console.log("Discount:", discount + "%");
```

- ◆ If cartTotal is 2000 or more → 20%
- ◆ Else, if member → 10%
- ◆ Else → 0%


```
js
let cartItems = 0;

if (cartItems === 0) {
  console.log("Cart is empty 🛒");
} else {
  console.log("Proceed to checkout ✅");
}
```

Copy code

✅ Task 3 Solution

Block discount if user is blacklisted

```
js
let cartTotal = 1500;
let isMember = true;
let hasCoupon = true;
let isBlacklisted = true;

if (!isBlacklisted && (cartTotal >= 1000 && isMember || hasCoupon)) {
  console.log("Discount applied 🎁");
} else if (isBlacklisted) {
  console.log("Discount blocked ❌ (User blacklisted)");
} else {
  console.log("No discount ❌");
}
```

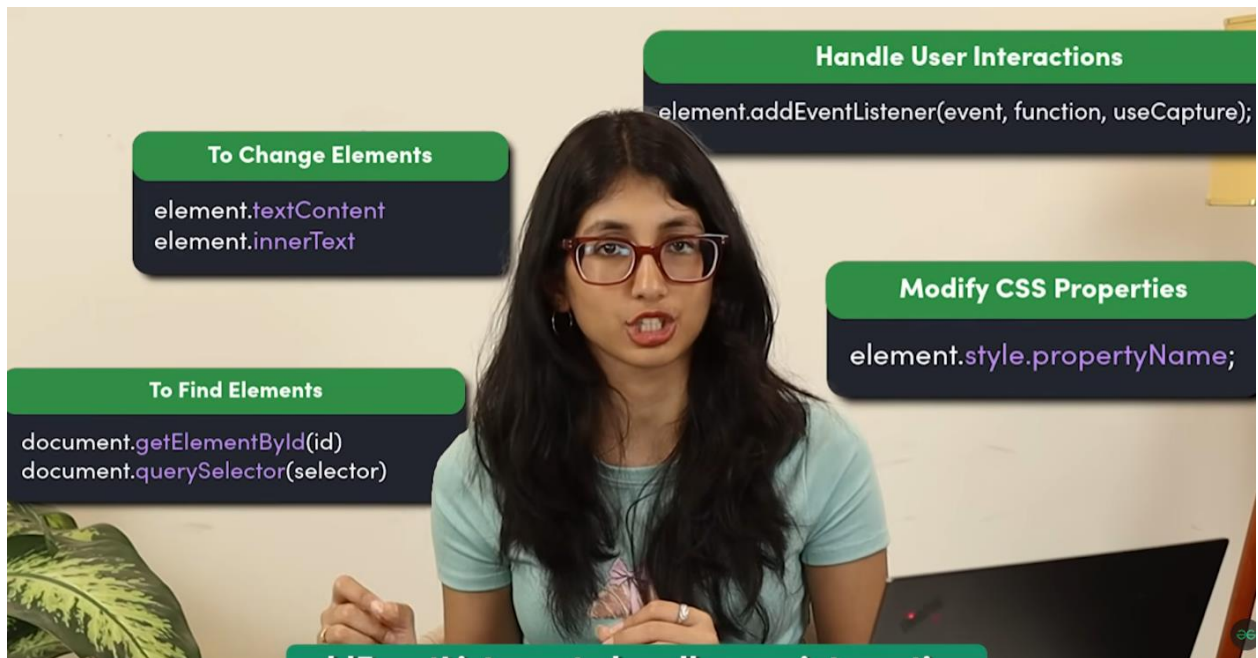
Copy code

```
let name = "Dipkiran";
let hour = new Date().getHours();

if (hour >= 5 && hour < 12){
  console.log ("Good morning," + name);
}
else if (hour >= 12 && hour < 17){
  console.log ("Good Afternoon," + name);
}

else if (hour >= 17 && hour < 21){
  console.log ("good Evening," + name);
}
```

}



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